Analysis of Algorithms: Lecture 2

MCQs with Answers · Complexity, Dynamic Programming, Matrix Chain

★ = **most likely to come.** All answers are taken from the Lecture 2 slides.

Background and complexity

1. An algorithm is a sequence of computational steps that transforms:

a) output into input
b) input into output
c) code into data
d) data into memory

Answer: b) input into output

2. Analysis of an algorithm finds its performance characteristics via:

a) cost and lines
b) time and space
c) input and output
d) compiler and OS

Answer: b) time and space

3. Algorithms are independent of: ★

a) input size
b) hardware, OS, compilers and languages
c) time
d) space

Answer: b) hardware, OS, compilers and languages

4. Which method of measuring complexity is independent of machine, OS, compiler and language? ★

a) Absolute value (bytes, seconds)
b) Size of the problem, n
c) Counting seconds
d) Counting bytes

Answer: b) Size of the problem, n

Space complexity

5. C[i] = A[i] + B[i] for arrays of size N needs how much space? ★

a) N
b) 2N
c) 3N
d) N^2

Answer: c) 3N (A, B and C each take N)

6. If we reuse B as B[i] = A[i] + B[i], the space needed is: ★

a) N
b) 2N
c) 3N
d) 4N

Answer: b) 2N

7. Many algorithms compromise space requirements because:

a) memory is costly
b) memory is cheap
c) time is cheap
d) space is unlimited

Answer: b) memory is cheap

Time complexity

8. Time complexity is considered as the number of: ★

a) seconds
b) basic operations
c) lines of code
d) bytes

Answer: b) basic operations (Arithmetic op, assignment, read, write)

9. Actual time in seconds is hard to measure because it:

a) is always zero
b) varies from machine to machine
c) is too large
d) needs Big-O

Answer: b) varies from machine to machine

10. For "Initialize n; For i = 0 to n; Print i", the total cost is: ★

a) n + 1
b) 2n + 4
c) n + 2
d) 3n

Answer: b) 2n + 4 (1 + (n+2) + (n+1))

11. In the above code, the line "For i = 0 to n" costs:

a) n
b) n + 1
c) n + 2
d) 2n

Answer: c) n + 2

12. Labels like Begin, End and End For cost:

a) 1
b) n
c) nothing
d) 2

Answer: c) nothing

13. T(n) = 2n + 4 is: ★

a) O(1)
b) O(n)
c) $O(n^2)$
d) O(log n)

Answer: b) O(n)

Big-O notation

14. f(n) is O(g(n)) if there exist constants k and N such that for all n ≥ N: ★

a) $f(n) \geq k \cdot g(n)$
b) $f(n) \leq k \cdot g(n)$
c) $f(n) = k \cdot g(n)$
d) $f(n) < g(n)$

Answer: b) $f(n) \leq k \cdot g(n)$

15. In f(n) = O(g(n)), g(n) is like a/an _____ for f(n).

a) lower bound
b) upper bound
c) exact value
d) average

Answer: b) upper bound

16. Big-O analysis is useful because we want our algorithms to be:

a) short
b) scalable
c) recursive
d) colourful

Answer: b) scalable (Predicts efficiency for large real-world inputs)

17. For T(n) = 7n + 100 and very large n, T(n) can be approximated by: ★

a) 100
b) 7n
c) 7n + 100
d) n^2

Answer: b) 7n (Ignore the 100 term)

18. For T(n) = 7n + 100, when n = 1 the contributing factor is:

a) 7n
b) 100
c) both equally
d) n^2

Answer: b) 100 (T(1) = 107)

19. T(n) = n^2 + 100n + log n + 1000 is approximated by: ★

a) 100n
b) 1000
c) n^2
d) log n

Answer: c) n^2 (Ignore the last 3 terms)

Growth rates

20. The statement sum++; is: ★

a) O(1)
b) O(n)
c) O(n^2)
d) O(log n)

Answer: a) O(1)

21. A single loop for (i=0; i<n; ++i) sum++; is: ★

a) O(1)
b) O(n)
c) O(n^2)
d) O(n^3)

Answer: b) O(n)

22. Two nested loops, each running n times, is: ★

a) O(n)
b) O(2n)
c) O(n^2)
d) O(log n)

Answer: c) O(n^2)

23. In the comparison table (one operation = 10^{-11} s), 2^N for N = 64 takes about:

a) 5 seconds
b) 5 hours
c) 5 years
d) 6 billion years

Answer: c) 5 years

24. For N = 8, N $\log_2$N equals:

a) 8
b) 16
c) 24
d) 64

Answer: c) 24

25. The growth rate of a function is also called: ★

a) exact time
b) asymptotic growth rate / order
c) memory size
d) compiler speed

Answer: b) asymptotic growth rate / order

Dynamic Programming

26. Dynamic Programming solves a problem by breaking it into subproblems and: ★

a) ignoring the results
b) storing the results of subproblems
c) solving each one many times
d) sorting them

Answer: b) storing the results of subproblems

27. The two main properties of a DP problem are: ★

a) Greedy choice and sorting
b) Overlapping subproblems and optimal substructure
c) Divide and conquer
d) Recursion and iteration

Answer: b) Overlapping subproblems and optimal substructure

28. DP is NOT useful when: ★

a) subproblems overlap
b) there are no common (overlapping) subproblems
c) the problem is large
d) there is a table

Answer: b) there are no common (overlapping) subproblems

29. In the recursion tree of fib(5), fib(3) is called: ★

a) 1 time
b) 2 times
c) 3 times
d) 5 times

Answer: b) 2 times

30. Memoization is a _____ approach. ★

a) Bottom-up
b) Top-down
c) Greedy
d) Random

Answer: b) Top-down

31. Tabulation is a _____ approach. ★

a) Top-down
b) Bottom-up
c) Recursive only
d) Heuristic

Answer: b) Bottom-up

32. In memoization, the lookup array is initialized with:

a) 0
b) NIL values
c) 1
d) random values

Answer: b) NIL values (NIL = -1 in the slides)

33. In which version is the table filled on demand? ★

a) Tabulated
b) Memoized
c) Both
d) None

Answer: b) Memoized

34. In which version are all entries of the table necessarily filled?

a) Memoized
b) Tabulated
c) Neither
d) Both

Answer: b) Tabulated

35. The tabulated program builds the table and returns:

a) the first entry
b) the last entry
c) the middle entry
d) the sum

Answer: b) the last entry

36. The tabulated Fibonacci program in the slides with n = 9 prints:

a) 21
b) 34
c) 55
d) 9

Answer: b) 34 (0,1,1,2,3,5,8,13,21,34)

Optimal substructure

37. When the optimal solution of a problem can be obtained from optimal solutions of its subproblems, it is called: ★

a) Overlapping subproblems
b) Optimal substructure
c) Memoization
d) Tabulation

Answer: b) Optimal substructure

38. Which problem has the optimal substructure property? ★

a) Longest simple path
b) Shortest path
c) Both
d) None

Answer: b) Shortest path

39. Which problem does NOT have the optimal substructure property? ★

a) Shortest path
b) Longest (simple) path
c) Matrix chain
d) LCS

Answer: b) Longest (simple) path

40. Floyd–Warshall and Bellman–Ford are typical examples of: ★

a) Greedy algorithms
b) Dynamic Programming
c) Divide and Conquer
d) Heuristics

Answer: b) Dynamic Programming (All pair shortest path)

41. In the slides, the longest path from q to r is:

a) q→r
b) q→s→t→r
c) q→t→r
d) q→s→r

Answer: b) q→s→t→r

Matrix Chain Multiplication

42. The MCM problem is to: ★

a) perform the multiplications
b) decide the order of multiplications
c) add matrices
d) find the inverse

Answer: b) decide the order of multiplications

43. Matrix multiplication is:

a) commutative
b) associative
c) not possible
d) random

Answer: b) associative ((ABC)D = (AB)(CD) = A(BCD))

44. A = 10×30, B = 30×5, C = 5×60. The cost of (AB)C is: ★

a) 4500
b) 27000
c) 9000
d) 1500

Answer: a) 4500 (1500 + 3000)

45. For the same matrices, the cost of A(BC) is: ★

a) 4500
b) 27000
c) 18000
d) 9000

Answer: b) 27000 (9000 + 18000)

46. Multiplying a (p × q) matrix with a (q × r) matrix costs:

a) p + q + r
b) p × q × r
c) p × r
d) q × r

Answer: b) p × q × r

47. In MCM, the i-th matrix A_i has dimension:

a) p[i] × p[i+1]
b) p[i-1] × p[i]
c) p[i] × p[i]
d) i × i

Answer: b) p[i-1] × p[i]

48. For p[] = {40, 20, 30, 10, 30}, the minimum number of multiplications is: ★

a) 24000 b) 26000
c) 30000 d) 36000

Answer: b) 26000 ((A(BC))D)

49. For p[] = {10, 20, 30, 40, 30}, the minimum number of multiplications is:

a) 26000 b) 30000
c) 6000 d) 24000

Answer: b) 30000 (((AB)C)D)

50. For p[] = {10, 20, 30}, the minimum number of multiplications is:

a) 600 b) 6000
c) 60000 d) 3000

Answer: b) 6000 (Only one way: 10×20×30)

51. In a chain of n matrices, the first set of parenthesis can be placed in:

a) n ways b) n − 1 ways
c) n + 1 ways d) 2n ways

Answer: b) n − 1 ways (ABCD: A(BCD), (AB)CD, (ABC)D)

52. Time complexity of the naive recursive MCM approach is: ★

a) O(n) b) $O(n^2)$
c) $O(n^3)$ d) Exponential

Answer: d) Exponential

53. Time complexity of the DP solution of MCM is: ★

a) O(n) b) $O(n^2)$
c) $O(n^3)$ d) $O(2^n)$

Answer: c) $O(n^3)$

54. In the MCM DP code, the cost of multiplying one matrix, m[i][i], is:

a) 1 b) 0
c) p[i] d) INT_MAX

Answer: b) 0

55. MCM DP builds the table m[][] in a _____ manner.

a) top-down b) bottom-up
c) random d) greedy

Answer: b) bottom-up

Short on time? Learn only the ★ MCQs

★ MCQ numbers: **3, 4, 5, 6, 8, 10, 13, 14, 17, 19, 20, 21, 22, 25, 26, 27, 28, 29, 30, 31, 33, 37, 38, 39, 40, 42, 44, 45, 48, 52, 53**

5 common traps in MCQs

1. **3N vs 2N**: C = A + B uses 3N; reusing B uses 2N.
2. **2n + 4**: 1 + (n + 2) + (n + 1). Labels cost nothing.
3. **Memoization = Top-down**, **Tabulation = Bottom-up**. Do not reverse them.
4. **Shortest path** has optimal substructure, **longest path does not**.
5. **(AB)C = 4500**, **A(BC) = 27000**. MCM DP = **$O(n^3)$**, naive = **exponential**.